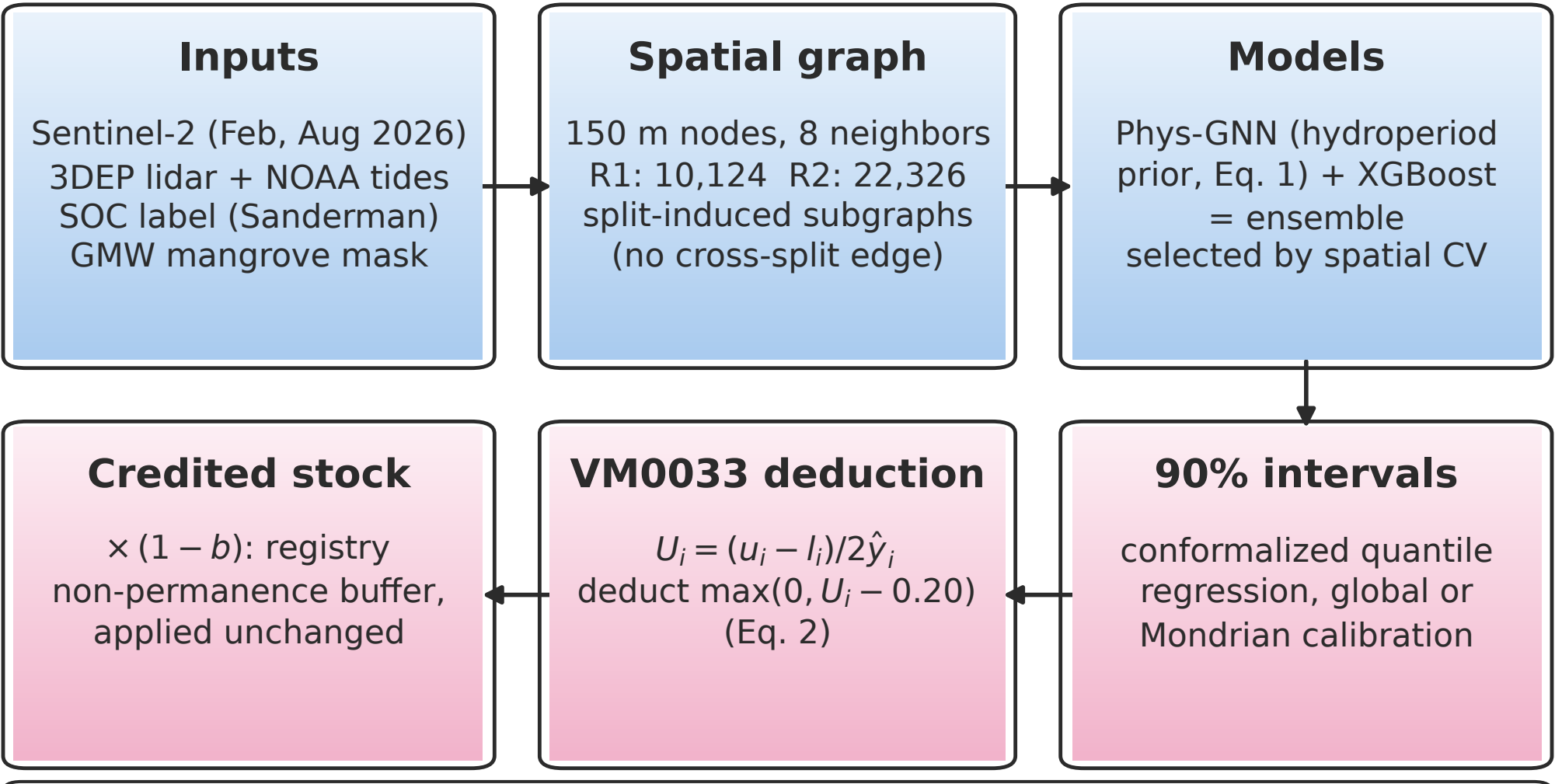

Inputs
Sentinel-2 (Feb, Aug 2026)
3DEP lidar + NOAA tides
SOC label (Sanderman)
GMW mangrove mask
Spatial graph
150 m nodes, 8 neighbors
R1: 10,124 R2: 22,326
split-induced subgraphs
(no cross-split edge)
Models
Phys-GNN (hydroperiod
prior, Eq. 1) + XGBoost
= ensemble
selected by spatial CV
90% intervals
conformalized quantile
regression, global or
Mondrian calibration
VM0033 deduction
$U_i = (u_i - l_i)/2\hat{y}_i$
deduct $\max(0, U_i - 0.20)$
(Eq. 2)
Credited stock
$\times (1 - b)$: registry
non-permanence buffer,
applied unchanged
Independent checks: 87 field cores (CCN), registry buffers (VROD), 2010-2020 mangrove loss